

# Module 3 - SQL Analysis Results

EN.605.256.82.FA26 Modern Software Concepts in Python  
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Final verified database snapshot: **30,015 rows**. This report contains all nine assigned questions plus two student-defined questions. Results are the final outputs after the successful Pull Data update. Percentages and averages are shown to two decimal places; counts are whole numbers.

## Q1. How many applicants are for Fall 2026?

### Final result

**29,535**

### SQL query used

```
SELECT COUNT(*)
FROM applicants
WHERE LOWER(TRIM(term)) = 'fall 2026';
```

### Explanation

Counts only rows whose source-backed term is explicitly Fall 2026.

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## Q2. What percentage of usable nationality classifications are international?

### Final result

**45.03%**

### SQL query used

```
SELECT
  100.0 * COUNT(*) FILTER (
    WHERE LOWER(TRIM(us_or_international)) = 'international'
  )
/ NULLIF(
  COUNT(*) FILTER (
    WHERE LOWER(TRIM(us_or_international)) IN (
      'international', 'american', 'other'
    )
  ),
  0
) AS international_percentage
FROM applicants;
```

### Explanation

International is the numerator. International, American, and Other form the usable denominator; missing/blank values are excluded.

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## Q3. What are the average GPA, GRE quantitative, GRE verbal, and GRE analytical-writing scores?

### Final result

**GPA: 3.77; GRE quantitative: 165.86; GRE verbal: 161.64; GRE analytical writing: 4.34**

### SQL query used

```
SELECT
  AVG(gpa) AS avg_gpa,
  AVG(gre) AS avg_gre_quantitative,
  AVG(gre_v) AS avg_gre_verbal,
  AVG(gre_aw) AS avg_gre_analytical_writing
FROM applicants;
```

### Explanation

PostgreSQL AVG ignores NULL independently for each metric, so one missing score does not remove a row from the other averages.

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## Q4. What is the average GPA of American applicants for Fall 2026?

### Final result

**3.78**

### SQL query used

```
SELECT AVG(gpa)
FROM applicants
WHERE LOWER(TRIM(term)) = 'fall 2026'
AND LOWER(TRIM(us_or_international)) = 'american';
```

### Explanation

Filters to explicit American and Fall 2026 classifications, then averages only non-NULL GPA values.

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## Q5. What percentage of Fall 2025 entries were accepted?

### Final result

**43.88%**

### SQL query used

```
SELECT
    100.0 * COUNT(*) FILTER (
        WHERE LOWER(TRIM(status)) LIKE 'accept%'
    )
    / NULLIF(COUNT(*), 0) AS accepted_percentage
FROM applicants
WHERE LOWER(TRIM(term)) = 'fall 2025';
```

### Explanation

Uses all explicit Fall 2025 entries as the denominator and accepted statuses as the numerator.

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## Q6. What is the average GPA of accepted Fall 2026 applicants?

### Final result

**3.76**

### SQL query used

```
SELECT AVG(gpa)
FROM applicants
WHERE LOWER(TRIM(term)) = 'fall 2026'
AND LOWER(TRIM(status)) LIKE 'accept%';
```

### Explanation

Filters to explicit Fall 2026 accepted entries and averages only non-NULL GPA values.

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## Q7. How many original downloaded records are Johns Hopkins master's applications in Computer Science?

### Final result

**7**

### SQL query used

```
SELECT COUNT(*)
FROM applicants
WHERE (
    LOWER(program) LIKE '%johns hopkins%'
    OR LOWER(program) ~ '^(^|^[a-z])jhu(^[a-z]|$)'
)
AND (
    LOWER(program) LIKE '%computer science%'
    OR LOWER(program) ~ '^(^|^[a-z])cs(^[a-z]|$)'
)
AND (
    LOWER(TRIM(degree)) LIKE 'master%'
    OR LOWER(TRIM(degree)) IN ('ms', 'm.s.', 'msc', 'm.sc.')
);
```

## Explanation

Uses only original/source-backed program and degree fields and recognizes Johns Hopkins/JHU plus Computer Science/CS and master's degree variants.

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## Q8. How many original-field Fall 2026 accepted PhD Computer Science records are at Georgetown, MIT, Stanford, or Carnegie Mellon?

### Final result

**25**

### SQL query used

```
SELECT COUNT(*)
FROM applicants
WHERE LOWER(TRIM(term)) = 'fall 2026'
  AND LOWER(TRIM(status)) LIKE 'accept%'
  AND (
    LOWER(TRIM(degree)) LIKE 'phd%'
    OR LOWER(TRIM(degree)) LIKE 'ph.d%'
    OR LOWER(TRIM(degree)) LIKE 'doctor%'
  )
  AND (
    LOWER(program) LIKE '%georgetown%'
    OR LOWER(program) LIKE '%massachusetts institute of technology%'
    OR LOWER(program) ~ '^(^|[a-z])mit([a-z]|$)'
    OR LOWER(program) LIKE '%stanford%'
    OR LOWER(program) LIKE '%carnegie mellon%'
    OR LOWER(program) ~ '^(^|[a-z])cmu([a-z]|$)'
  )
  AND (
    LOWER(program) LIKE '%computer science%'
    OR LOWER(program) ~ '^(^|[a-z])cs([a-z]|$)'
  );
```

## Explanation

Uses the original downloaded program field for institution/program matching plus source term, status, and degree fields.

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## Q9. Repeat Q8 using the LLM-generated program and university fields.

### Final result

**Original-field count: 25; LLM-field count: 25; Difference: +0**

### SQL query used

```
SELECT COUNT(*)
FROM applicants
WHERE LOWER(TRIM(term)) = 'fall 2026'
  AND LOWER(TRIM(status)) LIKE 'accept%'
  AND (
    LOWER(TRIM(degree)) LIKE 'phd%'
    OR LOWER(TRIM(degree)) LIKE 'ph.d%'
    OR LOWER(TRIM(degree)) LIKE 'doctor%'
  )
  AND (
    LOWER(llm_generated_university) LIKE '%georgetown%'
    OR LOWER(llm_generated_university) LIKE '%massachusetts institute of technology%'
    OR LOWER(llm_generated_university) ~ '^(^|[a-z])mit([a-z]|$)'
    OR LOWER(llm_generated_university) LIKE '%stanford%'
    OR LOWER(llm_generated_university) LIKE '%carnegie mellon%'
    OR LOWER(llm_generated_university) ~ '^(^|[a-z])cmu([a-z]|$)'
  )
  AND (
    LOWER(llm_generated_program) LIKE '%computer science%'
    OR LOWER(llm_generated_program) ~ '^(^|[a-z])cs([a-z]|$)'
  );
```

## Explanation

Uses the LLM-standardized program and university fields while retaining the source-backed term, status, and degree filters. The original-field and LLM-field totals are both 25 in this snapshot, so standardization did not change the total count. Equal totals do not prove that every matched row is identical, and normalization of abbreviations or institution/program names could change the matched set in another snapshot.

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## Q10. Own question: What percentage of all records with a usable status are accepted?

### Final result

**36.37%**

### SQL query used

```
SELECT
  100.0 * COUNT(*) FILTER (
    WHERE LOWER(TRIM(status)) LIKE 'accept%'
  )
  / NULLIF(COUNT(*), 0) AS accepted_percentage
FROM applicants
WHERE status IS NOT NULL
  AND TRIM(status) <> '';
```

## Explanation

This analysis uses the status field directly and calculates the share of records with a usable status that are classified as accepted.

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## Q11. Own question: Which five LLM-standardized universities have the most application records?

### Final result

**Stanford University: 667; Yale University: 601; University of California, Berkeley: 543; University of Wisconsin-Madison: 533; University of Chicago: 530**

### SQL query used

```
SELECT llm_generated_university, COUNT(*) AS application_count
FROM applicants
WHERE llm_generated_university IS NOT NULL
  AND TRIM(llm_generated_university) <> ''
GROUP BY llm_generated_university
ORDER BY application_count DESC, llm_generated_university ASC
LIMIT 5;
```

## Explanation

Groups the non-null LLM-standardized university field, counts records in each group, orders the groups from largest to smallest, and returns the top five. Because the four newly pulled records do not yet have LLM-standardized university values, those rows are excluded by the query rather than being assigned fabricated values.